HE2003 Econometrics I
Tutorial Question Set No.7
To be discussed on Oct 14 (Monday), 2024

***There is NO tutorial session on Oct 7, 2024.**

1. (a) For testing $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$ at the 5% significance level, compare the t-test procedure with the p-value procedure.
   (b) Explain the definition of a 95% confidence interval. Explain the trade-off between the 95% confidence interval and a confidence interval with a higher coverage probability such as the 99% confidence interval.
   (c) Explain the relationship between a t-test at the 5% significance level and a 95% confidence interval.
   (d) Explain the difference between homoskedasticity and heteroskedasticity.

2. Tennessee conducted an experiment in which kindergarten students were randomly assigned to "regular" and "small" classes and given standardized tests at the end of the year. Regular classes contained approximately 24 students, and small classes contained approximately 15 students. Let $SmallClass$ denote a binary variable equal to 1 if the student is assigned to a small class and equal to 0 otherwise. A regression of $TestScore$ on $SmallClass$ yields

$$\begin{aligned} TestScore &= \underset{(1.6)}{918.0} + \underset{(2.5)}{13.9} \times SmallClass, \quad R^2 = 0.01, \end{aligned} \tag{1}$$

   (a) Do small classes improve test scores? By how much?
   (b) Is the estimated effect of class size on test scores statistically significant? Carry out a test at the 5% level.
   (c) Construct a 95% confidence interval for the effect of $SmallClass$ on $TestScore$.

3. For the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, a test of the null hypothesis $H_0 : \beta_1 = 5$ using the t-statistic yields a p-value of 0.012.
   (a) Does the 99% confidence interval contain $\beta_1 = 5$? Explain. Does the 95% confidence interval contain $\beta_1 = 5$? Explain.
   (b) Suppose that the true value of β_1 is actually 10 and the standard error of $\hat{\beta}_1$ is equal to 1. Can you illustrate graphically the Type II error of the t-test with 5% significance level for $H_0 : \beta_1 = 5$? Is the Type II error smaller than 50%? Explain. [Hint: (1) remember the Type II error is accepting H_0 when H_0 is wrong, (2) we can draw the distribution of the t-statistic when the true value of β_1 is actually 10, and then find out the area corresponding to the Type II error.]

4. Let $Y_1, ..., Y_n$ be i.i.d. draws from a distribution with mean μ. A test of $H_0 : \mu = 10$ vs. $H_1 : \mu \neq 10$ using the t-test statistic yields a p-value of 0.07.

   (a) Does the 90% confidence interval contain $\mu = 10$? Explain.

   (b) Can you determine if $H_0 : \mu = 7$ is rejected by the (two-sided) t-test with 5% significance level? If yes, provide the reason. If no, provide the counter-example. [Hint: Is the t-statistic for $H_0 : \mu = 7$ on the left-hand-side or right-hand-side of the t-statistic for $H_0 : \mu = 10$?]